

JSW Data Engineer Interview Preparation (40 Questions Each for SQL, BigQuery, Tableau & Python)

SQL Interview Questions & Answers

Q1. What is SQL?

Answer: SQL is used to manage and query relational databases.

Q2. Difference between WHERE and HAVING?

Answer: WHERE filters rows before grouping while HAVING filters grouped data.

Q3. What is INNER JOIN?

Answer: Returns matching rows from both tables.

Q4. What is LEFT JOIN?

Answer: Returns all rows from left table and matching rows from right table.

Q5. What is RIGHT JOIN?

Answer: Returns all rows from right table and matching rows from left table.

Q6. What is FULL JOIN?

Answer: Returns all matching and non-matching rows from both tables.

Q7. What is Primary Key?

Answer: Uniquely identifies each row in a table.

Q8. What is Foreign Key?

Answer: Creates relationship between two tables.

Q9. What is normalization?

Answer: Reduces redundancy and improves structure.

Q10. What is denormalization?

Answer: Combines tables for faster reads.

Q11. Difference between DELETE and TRUNCATE?

Answer: DELETE removes selected rows, TRUNCATE removes all rows quickly.

Q12. What is DROP?

Answer: Deletes complete table structure.

Q13. What is indexing?

Answer: Improves query performance.

Q14. What is GROUP BY?

Answer: Groups similar rows for aggregate functions.

Q15. What is ORDER BY?

Answer: Sorts result set.

Q16. What is a subquery?

Answer: Query inside another query.

Q17. What is CTE?

Answer: Common Table Expression for readable queries.

Q18. What are aggregate functions?

Answer: Functions like SUM, AVG, COUNT.

Q19. Difference between UNION and UNION ALL?

Answer: UNION removes duplicates while UNION ALL keeps duplicates.

Q20. What is ROW_NUMBER()?

Answer: Assigns unique numbers to rows.

Q21. Difference between RANK and DENSE_RANK?

Answer: RANK skips numbers after tie while DENSE_RANK does not.

Q22. How to find duplicate rows?

Answer: Use GROUP BY and HAVING COUNT(*) > 1.

Q23. How to find second highest salary?

Answer: Use subquery or DENSE_RANK.

Q24. What is a view?

Answer: Virtual table based on SQL query.

Q25. What is stored procedure?

Answer: Reusable SQL block stored in database.

Q26. What is trigger?

Answer: Automatically executed SQL action.

Q27. What is ACID property?

Answer: Ensures reliable database transactions.

Q28. What is transaction?

Answer: Group of SQL operations executed together.

Q29. What is COMMIT?

Answer: Saves transaction permanently.

Q30. What is ROLLBACK?

Answer: Undoes transaction changes.

Q31. What is NULL?

Answer: Represents missing or unknown value.

Q32. Difference between CHAR and VARCHAR?

Answer: CHAR is fixed length, VARCHAR is variable length.

Q33. What is schema?

Answer: Logical structure of database.

Q34. What is query optimization?

Answer: Improving query speed and efficiency.

Q35. What is a clustered index?

Answer: Stores table data physically sorted.

Q36. What is non-clustered index?

Answer: Separate structure pointing to table rows.

Q37. How to optimize slow queries?

Answer: Use indexes and avoid unnecessary joins.

Q38. What is self join?

Answer: Joining table with itself.

Q39. What is cross join?

Answer: Returns Cartesian product of two tables.

Q40. Why SQL in Data Engineering?

Answer: Used for data querying, transformation and reporting.

BigQuery Interview Questions & Answers

Q1. What is BigQuery?

Answer: Google Cloud serverless data warehouse.

Q2. Why BigQuery is used?

Answer: For large-scale analytics and fast querying.

Q3. What is partitioning?

Answer: Splits tables into smaller partitions.

Q4. What is clustering?

Answer: Organizes data based on columns.

Q5. What is ETL?

Answer: Extract, Transform and Load process.

Q6. What is ELT?

Answer: Load data first then transform later.

Q7. What is OLTP?

Answer: System for transactional processing.

Q8. What is OLAP?

Answer: System for analytical processing.

Q9. What is a Data Warehouse?

Answer: Centralized storage for analytics.

Q10. What is Star Schema?

Answer: Fact table connected to dimension tables.

Q11. What is Snowflake Schema?

Answer: Normalized version of Star Schema.

Q12. What are fact tables?

Answer: Contain measurable data.

Q13. What are dimension tables?

Answer: Contain descriptive attributes.

Q14. How to reduce BigQuery cost?

Answer: Avoid SELECT * and use partitions.

Q15. What is streaming insert?

Answer: Real-time insertion into BigQuery.

Q16. What is data ingestion?

Answer: Importing data into systems.

Q17. What is schema evolution?

Answer: Updating schema without breaking data.

Q18. What are scheduled queries?

Answer: Queries running automatically.

Q19. What is materialized view?

Answer: Precomputed query result for performance.

Q20. What is data pipeline?

Answer: Automated flow of data processing.

Q21. What is columnar storage?

Answer: Stores data column-wise for analytics.

Q22. What is denormalization in warehouse?

Answer: Combining tables for faster querying.

Q23. What is data lake?

Answer: Storage for raw structured and unstructured data.

Q24. Difference between Data Lake and Warehouse?

Answer: Warehouse is structured, lake stores raw data.

Q25. What is batch processing?

Answer: Processing large data in intervals.

Q26. What is real-time processing?

Answer: Processing data instantly.

Q27. What is Airflow?

Answer: Workflow orchestration tool.

Q28. What is Kafka?

Answer: Distributed event streaming platform.

Q29. What is Pub/Sub?

Answer: Google messaging service.

Q30. What is data governance?

Answer: Managing data quality and security.

Q31. What is metadata?

Answer: Data about data.

Q32. What is CDC?

Answer: Change Data Capture tracks updates.

Q33. What is data partition pruning?

Answer: Reading only required partitions.

Q34. How does BigQuery improve scalability?

Answer: Automatically scales resources.

Q35. What is SQL dialect in BigQuery?

Answer: Uses standard SQL.

Q36. What are nested fields?

Answer: Complex hierarchical data fields.

Q37. What is sharding?

Answer: Splitting data into multiple tables.

Q38. What is query caching?

Answer: Stores previous query results.

Q39. How to improve BigQuery performance?

Answer: Use clustering and optimized filters.

Q40. Why BigQuery for Data Engineering?

Answer: Fast, scalable and serverless analytics.

Tableau Interview Questions & Answers

Q1. What is Tableau?

Answer: Business intelligence and visualization tool.

Q2. What is a dashboard?

Answer: Collection of multiple visualizations.

Q3. What is worksheet?

Answer: Single chart or graph page.

Q4. Difference between dimension and measure?

Answer: Dimensions are categorical while measures are numerical.

Q5. What is a calculated field?

Answer: Custom formula inside Tableau.

Q6. What is parameter?

Answer: User input for dynamic dashboards.

Q7. What is filter?

Answer: Limits displayed data.

Q8. What is context filter?

Answer: Primary filter improving performance.

Q9. What is data blending?

Answer: Combining multiple data sources.

Q10. What is data joining?

Answer: Connecting tables using keys.

Q11. What is live connection?

Answer: Real-time connection to source.

Q12. What is extract connection?

Answer: Snapshot of data stored locally.

Q13. What is Tableau Server?

Answer: Platform for sharing dashboards.

Q14. What is Tableau Public?

Answer: Free platform to publish dashboards.

Q15. What is Tableau Prep?

Answer: Tool for data cleaning.

Q16. What is hierarchy?

Answer: Allows drill-down analysis.

Q17. What is heatmap?

Answer: Color-based data visualization.

Q18. What is KPI dashboard?

Answer: Displays key business metrics.

Q19. What is dual axis chart?

Answer: Two measures on same chart.

Q20. What is a story?

Answer: Sequence of dashboards for presentation.

Q21. How to improve dashboard performance?

Answer: Use extracts and reduce filters.

Q22. What is a data source filter?

Answer: Filter applied directly on source.

Q23. What is row-level security?

Answer: Restricts data visibility by user.

Q24. What is a gantt chart?

Answer: Displays project timelines.

Q25. What is map visualization?

Answer: Displays geographical data.

Q26. What is tooltip?

Answer: Extra information on hover.

Q27. What are sets?

Answer: Custom subset of data.

Q28. What is trend line?

Answer: Shows direction of data.

Q29. What is forecasting?

Answer: Predicting future values.

Q30. What is data storytelling?

Answer: Presenting insights visually.

Q31. What is blending vs joining?

Answer: Joining combines physically, blending combines logically.

Q32. What are actions in Tableau?

Answer: Interactive dashboard controls.

Q33. What is drill down?

Answer: Viewing detailed level data.

Q34. What is dashboard layout?

Answer: Arrangement of dashboard objects.

Q35. What is aggregation?

Answer: Summarizing data values.

Q36. What is Tableau Reader?

Answer: Tool to open Tableau files offline.

Q37. How to connect Tableau with SQL?

Answer: Using database connectors.

Q38. Why Tableau in Data Engineering?

Answer: Used for reporting and analytics.

Q39. What is calculated KPI?

Answer: Metric calculated using formulas.

Q40. Difference between dashboard and story?

Answer: Dashboard shows visuals while story explains insights.

Python Interview Questions & Answers

Q1. What is Python?

Answer: High-level programming language used in data engineering.

Q2. Difference between list and tuple?

Answer: List is mutable, tuple is immutable.

Q3. What is dictionary?

Answer: Key-value pair data structure.

Q4. What is set?

Answer: Unordered collection of unique values.

Q5. What is Pandas?

Answer: Library for data analysis.

Q6. What is NumPy?

Answer: Library for numerical operations.

Q7. What is DataFrame?

Answer: Tabular data structure in Pandas.

Q8. How to read CSV file?

Answer: Use `pandas.read_csv()`.

Q9. How to handle null values?

Answer: Use `fillna()` or `dropna()`.

Q10. What is lambda function?

Answer: Anonymous one-line function.

Q11. What is list comprehension?

Answer: Short syntax to create lists.

Q12. What is exception handling?

Answer: Handling errors using try-except.

Q13. What is OOP?

Answer: Programming using classes and objects.

Q14. What is inheritance?

Answer: Child class acquiring parent properties.

Q15. What is polymorphism?

Answer: Same method behaving differently.

Q16. What is encapsulation?

Answer: Wrapping data and methods together.

Q17. What is multithreading?

Answer: Running multiple threads simultaneously.

Q18. What is multiprocessing?

Answer: Running multiple processes together.

Q19. What is JSON?

Answer: Data exchange format.

Q20. How to call API in Python?

Answer: Using requests library.

Q21. What is file handling?

Answer: Reading and writing files.

Q22. What is virtual environment?

Answer: Isolated Python environment.

Q23. What is pip?

Answer: Python package installer.

Q24. What is generator?

Answer: Function yielding values one by one.

Q25. What is iterator?

Answer: Object used for iteration.

Q26. Difference between append and extend?

Answer: Append adds single item, extend adds multiple items.

Q27. What is slicing?

Answer: Extracting parts of sequence.

Q28. What is recursion?

Answer: Function calling itself.

Q29. What is deepcopy?

Answer: Copies nested objects completely.

Q30. How to connect Python with SQL?

Answer: Using sqlalchemy or psycopg2.

Q31. What is logging?

Answer: Tracking events in applications.

Q32. What is regex?

Answer: Pattern matching using regular expressions.

Q33. What is API?

Answer: Interface for communication between systems.

Q34. What is Flask?

Answer: Lightweight Python web framework.

Q35. What is FastAPI?

Answer: Modern high-performance API framework.

Q36. What is threading issue?

Answer: Concurrency problem due to shared resources.

Q37. What is GIL?

Answer: Global Interpreter Lock in Python.

Q38. Why Python for ETL?

Answer: Easy automation and data processing.

Q39. Why Python in Data Engineering?

Answer: Supports pipelines, APIs and analytics.

Q40. How to improve Python performance?

Answer: Use optimized libraries and efficient algorithms.